Reading Robustness from One Truncated Trace: Methods for Detecting Spurious Math Reasoning

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Abstract

The AIMO Interpretability Challenge asks whether a reasoning model's success on a competition math problem survives faithful rewordings of the problem. We predict this from a single greedy reasoning trace truncated at 6144 tokens, with white-box access to the model. Our method calls every problem whose reasoning finishes within the budget robust, and scores unfinished traces with separate methods per model: a rank ensemble of self-verification share, answer survival when the reasoning is cut, hidden-state sensitivity to programmatic rewordings, attention look-back and forced-answer confidence. for Qwen3.5-4B, we also use the entropy of the forced answer after 4096 tokens, and for Olmo-3-7B-Think, a Min-K% familiarity router and a second model acting as a judge. Our two submissions differ only in Olmo's unfinished calls: the second replaces them with an early-trace hidden-state direction. They score 1.00 and 0.93 on the public leaderboard (all-robust: 0.50) and make 33 and 36 correct calls on the official training labels for our four models (43 rows; all-robust: 31). Removing the output-distribution signals lowers leaderboard accuracy to 0.71–0.79; removing the hidden-state and attention signals leaves the first submission unchanged, while the second's hidden-state direction loses three training set points, while losing one on the leaderboard. We also report a negative result: labels we generated with our own perturbations do not reproduce the official labels, which shaped how we validated everything. Code: `https://anonymous.4open.science/r/AIMO-Interp-Challenge-Submission-Anonymised/`.

1 Introduction

Both nominated submissions share one test-time pipeline (Section 3) and differ only in how unfinished Olmo problems are called. Both methods are covered here. S1 (`permodel-v8-ent4096`) is the base method. S2 (`permodel-v8-ent4096-olmoseg25`) is S1 with one single-feature override for Olmo (Section 3.5).

Contributions. A pipeline that given a model and a problem, generates one greedy trace and a few teacher forced passes, and predicts whether the model was solving the problem robustly or not. We also include ablation studies and our reports on the various white-box methods we tried on this task

2 Data and evaluation

Models. Qwen3.5-4B [Qwen Team, 2026], Skywork-OR1-Math-7B [He et al., 2025a], Olmo-3-7B-Think [Team Olmo, 2025] and DeepSeek-R1-0528-Qwen3-8B [DeepSeek-AI, 2025, Yang et al., 2025]; perturbation robustness in the sense of Mirzadeh et al. [2025], labelled by the challenge [Stefánik et al., 2026].

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Official labels. MAIN: the organisers' training set `train-main-v2`, restricted to our four models: 43 rows (Qwen 12, Skywork 12, Olmo 9, DeepSeek 10), 31 robust. This is the only labelled data we evaluate on; we report correct calls out of 43 with Wilson 95% intervals [Wilson, 1927]. **Public leaderboard.** We report accuracy on the leaderboard's 14 problems. With 3-5 problems per model its estimates are also very noisy.

Our own data, used for calibration only. Unlabelled greedy traces for $\sim$100 post-cutoff competition problems per model (POOL) supply the reference distributions and median thresholds below, which need no labels. We also labelled problems ourselves (released with the code). No official threshold for the accuracy drop was published, so we inferred the rule from the data. In MAIN, every robust row has a worst-case drop ≤ 0.10 and every non-robust row ≥ 0.25, with nothing in between, and the same rule, counting never-solved problems as robust, reproduces the labels of the public validation sample exactly. We used 8 temperature-1 samples per version and five perturbation types, four written by Claude Sonnet and one (typos) scripted. Every generation of our own was capped at 6144 new tokens, matching the test-time trace, with answers forced at the cap when the model had not finished (the relabel in Section 6 used 32k); the official labels come from a much larger generation budget. These labels are used only for low-capacity choices: the sign of each feature, the per-model feature subsets and, for S2, one direction. Section 6 shows that they are noisy and track difficulty, so every such choice was kept only if it also held on the official labels, and no accuracy below is measured on our labels (the "ours" column of Table 4 is a diagnostic).

Regime split. At a 6144-token cap most MAIN traces are still thinking (32/43). All 11 MAIN rows that finish are robust. Every non-robust label therefore falls on an unfinished trace, which is where every recipe acts.

3 Method

Trace. We build the challenge prompt ("Solve the following mathematical problem step by step. Put your final answer within `\boxed{}`") with each model's chat template. We decode greedily (bf16, HF Transformers [Wolf et al., 2020]) up to 6144 new tokens, stopping early on end-of-thinking, end-of-sequence or a 3$\times$32-token loop. **Finished** (the end-of-thinking token appears) $\Rightarrow$ robust. **Unfinished** $\Rightarrow$ score with the model's recipe.

All features are computed on that one trace (definitions in Appendix A). We group them by what they read; Section 5.1 ablates the groups.

3.1 Text features (black-box)

Finished rule: whether the end-of-thinking token appears within the cap. `chk`: the share of $\geq$4-word sentences of the reasoning that verify or check something.

3.2 Output-distribution features (grey-box)

These read the model's next-token distribution under teacher forcing. `fa`: the top-1 probability of the first answer token after appending "`</think>` The final answer is `$\boxed{`" at 6144 tokens. H_b: the entropy of that distribution when the trace is cut at b tokens. *Answer survival* [cf. early answering, Lanham et al., 2023]: `nrk` and `f1k`, the probability of the clean forced answer when the reasoning is removed entirely or cut to its first 1024 tokens, relative to the clean probability. `cfl`: the change in the trace's log-likelihood when it is teacher-forced after three automatic rewordings of the problem (10% typos, a neutral distractor sentence, renamed variables). Min-K% [Shi et al., 2024] ($k = 20$): the mean of the lowest 20% token log-probabilities of the problem text, it is a familiarity score.

3.3 Internal features (white-box)

Hidden states: `cfc`, the cosine between late-trace hidden states of the original pass and of the three reworded passes; the Olmo direction of S2 (Section 3.5). *Attention*: `lble` and `pmle`, the Lookback-Lens ratio [Chuang et al., 2024] and the attention mass on the prompt, each as last quarter minus first quarter of the trace; `jmp`, the attention-weighted mean backward distance between 64-token blocks (robust traces keep re-reading their own setup). The Qwen judge used for Olmo (below) is also internal, since Qwen's ensemble consists of `lble`, `pmle`, `jmp` and `cfc`.

Table 1: Correct calls on MAIN (43 rows) with Wilson 95% intervals, and public leaderboard accuracy (14 problems). On MAIN, Olmo's judge needs a Qwen trace of each Olmo problem, which exists for 2 of the 4 unfinished Olmo rows; there it makes the same calls as the router, so Olmo's S1 calls use the router branch. S1 is best on the leaderboard, S2 on MAIN; on MAIN both gains over all-robust are inside the intervals.

Method	MAIN correct (43)	Leaderboard score
All robust (baseline)	31 [0.57, 0.83]	0.50
Finished ⇒ R, unfinished ⇒ NR	23 [0.39, 0.67]	0.57
S1 (`v8-ent4096`)	33 [0.62, 0.87]	**1.00**
S2 (S1 + Olmo direction)	**36** [0.70, 0.92]	0.93

3.4 From features to calls

Rank-ensemble score. For model m with feature subset F_m, a value x_f is mapped to its percentile q_f among the reference values $R_{m,f}$ (POOL traces plus our labelled traces, unfinished only):

$$s = \frac{1}{|F_m|} \sum_{f \in F_m} \left(\tilde{q}_f - \tfrac{1}{2}\right), \qquad \tilde{q}_f = \begin{cases} q_f & \sigma_f = +1 \\ 1 - q_f & \sigma_f = -1 \end{cases}, \qquad \text{robust} \iff s > \tau_m. \tag{1}$$

τ_m is the median score s of model m's calibration traces. The signs σ_f are the direction of each feature's length-controlled AUC. The subsets F_m were chosen per model by subset search on our labelled rows: Qwen {`lble, pmle, jmp, cfc`} (Qwen's own unfinished calls use H_{4096} instead, below; this ensemble serves as the judge for Olmo); Skywork {`fa, chk, f1k`}; DeepSeek {`chk, nrk, cfl, pmle`}. Percentile scoring makes the score insensitive to feature scale and to shifts between our hardware and the evaluator's.

Olmo: novelty router and judge. Olmo's features flipped direction between problems it had seen in training and post-cutoff problems. We route on Min-K%: below -5.93 (the cut that best separated post-cutoff from older problems in POOL, using no robustness labels) goes to a novel branch {`f1k`}, otherwise to a familiar branch {`chk, lble, pmle`}, each with its own median threshold. A judge then loads Qwen, generates Qwen's own trace of the same problem and scores it with Qwen's recipe. Olmo's call is robust iff the mean of two label-free percentiles exceeds 0.5: Olmo's margin among its POOL margins, and Qwen's margin among Qwen's margins on the same POOL problems. If the judge fails or time is short, Olmo's own call stands.

Qwen: forced-answer entropy at 4096 tokens. For unfinished Qwen problems, S1 replaces the rank-ensemble call with H_{4096} (applied last, after any judging): robust iff $H_{4096} < 1.621$, the median over Qwen's 100 POOL traces. The sign comes from our 66 labelled Qwen problems (AUC 0.77 for lower-is-robust). Qwen's rank-ensemble margins are still computed, because the Olmo judge uses them.

3.5 S2: an early-trace hidden-state direction for Olmo

S2 runs the full S1 pipeline and then, after the judge, replaces the call only for unfinished Olmo problems. The router and judge are only a fallback for Olmo in S2. We take the mean hidden state over the 256 tokens around 25% of the trace, at the 5th decoder layer, and project it onto a precomputed unit robust-minus-spurious difference-of-means direction fitted on our 21 labelled unfinished Olmo problems. Robust iff the projection is above 0.234, the median over Olmo's 50 POOL traces. For both single-feature readouts the threshold is label-free and the sign comes from our data. Choosing which feature to use for these two models, however, used MAIN results and leaderboard scores of earlier builds (see Limitations).

4 Results

Table 1 gives the headline numbers. On the leaderboard S1 scores 1.00 and S2 0.93, against 0.50 for all-robust. Both leaderboard scores are upper bounds rather than estimates of private-set accuracy: the recipes, including S1's Qwen rule and S2's Olmo direction, were chosen with the leaderboard scores of earlier builds in view. MAIN is the less biased measure, and there the gains are small: it

Table 2: Component ablation: correct calls per model on MAIN (Qwen 12, Skywork 12, Olmo 9, DeepSeek 10 rows; of them 11/7/4/10 unfinished) and leaderboard accuracy (LB). The per-model recipes help DeepSeek and Skywork most; the Qwen entropy helps on both sets; the Olmo direction helps on MAIN but costs one leaderboard problem.

Variant	Qwen	Skywork	Olmo	DeepSeek	Total	LB
All robust	8	9	7	7	31	0.50
Finished only (unfinished NR)	5	8	7	3	23	0.57
v8 (rank ensembles, no overrides)	8	10	5	9	32	0.93
v8 without Olmo router/judge	8	10	7	9	34	0.79
S1 = v8 + Qwen H_{4096}	9	10	5	9	33	1.00
v8 + Olmo direction only	8	10	8	9	35	0.86
S2 = S1 + Olmo direction	9	10	8	9	36	0.93

Table 3: White-box ablation by feature group. *No internals*: every hidden-state and attention feature, the Olmo direction and the Qwen judge removed; the two submissions then coincide, since they differ only in an internal readout. *No output*: `fa`, `nrk`, `f1k`, `cfl`, the Min-K% router and H_{4096} removed (Qwen falls back to its internal ensemble). *Text only*: finished rule + `chk` for every model. A model left with no feature falls back to `chk`. Modified recipes use threshold 0 (S1 at threshold 0: 32 / 1.00). Output features carry most of the signal; of the internal readouts, only S2's Olmo direction changes the totals.

Variant	Qwen	Skywork	Olmo	DeepSeek	MAIN	LB
All robust	8	9	7	7	31	0.50
Text only	10	9	8	7	34	0.57
Text + output (no internals; S1 = S2)	9	10	6	8	33	1.00
S1: text + internal (no output)	8	9	6	7	30	0.71
S2: text + internal (no output)	8	9	8	7	32	0.79
S1 (all three)	9	10	5	9	33	1.00
S2 (all three)	9	10	8	9	36	0.93

is 72% robust, S2 beats all-robust by 5 calls and S1 by 2, within the intervals, and the text-only variant (finished rule and check share) already reaches 34/43 (Table 3). The two submissions trade places across the two label sets, which is why we nominate both. Among all builds we scored on the leaderboard (Appendix E), the per-model builds scored 0.64–1.00 and single-family builds 0.28–0.71.

5 Ablations

Components (Table 2). The regime split is not a predictor on its own but settles 11/43 MAIN rows at zero error. The Olmo router trades two MAIN rows for two leaderboard problems; the judge makes the same leaderboard calls as the router. S1 keeps the router and judge for Olmo and is the hedge against the Olmo direction of S2, which rests on only 4 unfinished MAIN rows.

5.1 Does the white-box signal matter?

The internal signals are DeepSeek's prompt-attention feature in both submissions, plus Olmo's attention-based familiar branch and Qwen judge in S1 and Olmo's hidden-state direction in S2; Qwen's own calls use H_{4096}, an output feature, in both. In S1 the internals make no measurable difference: without them the recipe also scores 33/43 and 1.00, because Olmo's unfinished calls are decided mostly by the Min-K% router, an output feature, and the attention-based familiar branch calls almost every problem it receives robust. The internal readout that matters is S2's hidden-state direction, which adds 3 MAIN rows for one leaderboard problem and gives the best total over both sets (49/57, against 47 without internals). Output features carry most of the signal: without them leaderboard accuracy falls to 0.71–0.79. Recalibrating every variant's thresholds label-free on the unlabelled pool moves individual cells by up to 3 calls but leaves these conclusions unchanged. Appendix B removes one feature at a time.

Table 4: Where in the trace to read the signal: AUC on unfinished traces (robust = positive class, oriented by the shipped sign) on our labels / MAIN. Left: Qwen forced-answer entropy at truncation b (MAIN: 7 R / 4 NR). Right: Olmo direction projections by trace position (MAIN: 2 R / 2 NR). Olmo's "ours" AUCs are in-sample, since the direction was fitted on those labels. On our labels the entropy signal grows with b; on MAIN every budget is similar, so the exact budget matters little.

Qwen H_b, $b =$	ours	MAIN
512	0.44	0.75
1024	0.49	0.64
2048	0.61	0.71
3072	0.69	0.75
4096	0.77	0.71
6144	0.79	–

Olmo direction at	ours	MAIN
25% of trace	0.99	1.00
50% of trace	0.97	1.00
75% of trace	0.93	1.00

Truncation point and position (Table 4). On our labels the forced-answer entropy becomes informative only after about 2048 tokens of reasoning; on MAIN all budgets give 0.64–0.75, so the choice of 4096 (made with MAIN and earlier leaderboard scores in view) matters little. For Olmo, the direction separates MAIN's four unfinished rows at every position, so 25% is not supported over 50% or 75%; this is the weakest-supported component, and the reason S1 is nominated alongside S2.

Alternatives tried (Appendices D and E). We also tested whole-trace hidden-state geometry (K-means walk graphs, δ-hyperbolicity and Poincaré embeddings [Raj, 2026], CKA [Kornblith et al., 2019]), attention-map spectra [Bińkowski et al., 2025], Min-K%++ [Zhang et al., 2025], kNN and TDA trajectory features, PEARL-style prompt noising, and sentence-embedding reasoning graphs. Several reach high AUC on one label set but none holds up across our labels and MAIN; builds based on single families scored 0.28–0.71 on the leaderboard.

6 What do the official labels measure?

We tried to enlarge the labelled data by applying the official rule ourselves, and found two problems with the resulting labels.

Noise and difficulty. With 8 samples per version, a problem solved about half the time can cross the 0.25 drop threshold by chance. Our first labelled problems were also much harder for the models than MAIN's: on unfinished traces, the median forced-answer top-1 probability at 6144 tokens was 0.54 against 1.00 for Qwen, and 0.92 against 1.00 for Skywork. We then labelled a second set of 109 problems matched to MAIN's confidence (forced top-1 probability at least 0.7). It closed the difficulty gap but not the noise: we obtained 10, 10 and 8 non-robust labels for Qwen, Olmo and Skywork, about as many as a no-effect null predicts (9.3, 10.3 and 7.1). Relabelling 62 Skywork problems at 32k with natural answers left 8 non-robust cases; our Skywork recipe was at chance on them (balanced accuracy 0.52), and original-problem accuracy still predicted the label (AUC 0.74).

Non-reproduction. Our perturbations also miss official non-robust cases. MAIN problem `7ee9d443` is officially non-robust for all four models (drops 0.40–1.00), yet under our perturbations Qwen and Skywork solve it in 47–48 of 48 samples at a 32k cap, in 67–68 of 68 across 30 further rewordings, and also with changed numbers and other prompt formats. Our sampling settings match the official ones, so the official labels must depend on perturbation texts or grading details we do not have.

Consequence. Such labels can still fix the sign of a consistent feature, but cannot measure accuracy. We therefore evaluate only on MAIN and the leaderboard, and keep each choice made on our labels (Section 2) only if it also holds on the official ones.

Limitations

Small, partly in-sample evaluation. MAIN's 43 rows give intervals about ± 0.12 wide; S1 and S2 differ by 3 MAIN calls and one leaderboard problem, neither of which is significant. MAIN is not fully held out: feature signs for the shared features used MAIN (length-controlled AUC), reference sets include MAIN's unfinished values, and both single-feature readouts, as well as the per-model subsets of later builds, were chosen using MAIN results and the leaderboard scores of earlier builds.

Unreliable self-generated labels (Section 6) leave the signs and the Olmo direction resting on 21–66 noisy labels per model. **Distribution shift.** Features calibrated on our data transferred unevenly to leaderboard scores (single-family builds 0.28–0.71), so accuracy on the private 43 problems may differ substantially from MAIN. **Replay gap.** Olmo's judge was replayed on only 2 of the 4 unfinished MAIN rows. **Greedy-only view.** We observe one greedy trace capped at 6144 tokens, while the labels come from temperature-1 samples with a much larger budget.

AI usage statement

We used AI assistants extensively. **Claude Code, mainly with Claude Opus 5.5 (Anthropic),** was used for much of the code and experimentation: it wrote most of the analysis, feature-extraction, labelling and submission code under our direction; ran and monitored the GPU experiments. **Claude Sonnet** (through claude.ai) generated the perturbation texts for our own labelled data, following a written specification (Section 2). **Gemini Deep Research** (Google) and **Claude Sonnet** assisted the literature review, suggesting candidate papers and methods; we read and checked every paper we used. The authors chose the research directions, decided every submission and nomination, checked the reported numbers against the experiment logs, and are responsible for all claims. No AI system is an author.

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A Feature definitions

All features are computed on the prompt p and the greedy trace t (at most 6144 tokens), with suffix σ = "`\n</think>\n\nThe final answer is $\boxed{`".

- `chk`: share of the thinking's sentence units ($\geq$4 words) that match a check/verify/confirm lexicon; undefined below 10 units.
- `fa`: $\max_v P(v \mid p, t_{1:6144}, \sigma)$.
- `nrk`, `f1k`: with a^* the clean forced answer token, $P(a^* \mid p, \sigma)/P(a^* \mid p, t, \sigma)$ and $P(a^* \mid p, t_{1:1024}, \sigma)/P(a^* \mid p, t, \sigma)$.
- `cfc`, `cfl`: for each automatic rewording p' (typos at rate 0.1, a fixed distractor sentence after the first sentence, single-letter variables in math renamed consistently), teacher-force t after p'. `cfc` is the mean cosine between original and reworded 16-token-window hidden states over the last quarter (averaged over 6 layers); `cfl` is the mean change in per-token log-likelihood of t over its first tokens.
- `lble`, `pmle`: per 16-token span, all layers and heads, the Lookback-Lens ratio (attention on the prompt over attention on prompt plus generated tokens) and the raw prompt attention mass; mean over the last quarter of spans minus the first quarter.
- `jmp`: on the 64-token block graph (each block's attention on earlier blocks, normalised per block), the mean attention-weighted backward distance in blocks.
- H_b: entropy of $P(\cdot \mid p, t_{1:b}, \sigma)$.
- seg-25% direction: $\langle \bar{h}^{(5)}_{[0.25L \pm 128]}, d \rangle$, with $\bar{h}^{(5)}$ the mean 5th-layer state over the 256 tokens around position $0.25L$, and d the normalised difference of class means on our labelled unfinished Olmo traces.
- Min-K% ($k = 20$): mean of the lowest 20% token log-probabilities of the problem text under the model.

B Drop-one-feature ablation

Table 5 removes one feature at a time from S1's recipe, keeping the shipped thresholds; for Olmo, features are removed from the router branch that uses them. Scores are correct calls on MAIN (43 rows) and the leaderboard (14).

Table 5: Drop-one-feature ablation of S1 (33 / 14). Group: T = text, O = output distribution, I = internal. Every Skywork and DeepSeek feature carries weight, including DeepSeek's prompt-attention feature (2 MAIN rows); removing any feature from Olmo's familiar branch gains one MAIN row at no leaderboard cost.

Model	Feature removed	Group	MAIN (43)	LB (14)
—	none (S1)	—	33	14
Skywork	`fa`	O	32	14
Skywork	`chk`	T	33	13
Skywork	`f1k`	O	33	12
DeepSeek	`chk`	T	32	13
DeepSeek	`nrk`	O	31	14
DeepSeek	`cfl`	O	31	14
DeepSeek	`pmle`	I	31	14
Olmo, familiar branch	`chk`	T	34	14
Olmo, familiar branch	`lble`	I	34	14
Olmo, familiar branch	`pmle`	I	34	14
Olmo, novel branch	`f1k` (replaced by `chk`)	O	35	12
Qwen	H_{4096} (back to the rank ensemble)	O	32	13

C Per-feature evidence

Table 6 gives each shipped feature's own AUC on unfinished traces, oriented by the sign the recipe uses, so values above 0.5 mean the feature points the way it is used. The columns are our first labelled problems (the signs were chosen there, so these are in-sample), a second labelled set that we drew to match MAIN's forced-answer confidence and used for no choice, and MAIN. All use forced labels at 6144 tokens, and no length control is applied. The MAIN counts are small (Olmo 2 + 2), so a single row moves an AUC by up to 0.25. On the first labelled set, every feature except Skywork's

Table 6: AUC of each shipped feature on unfinished traces, oriented by its shipped sign. n = robust / non-robust traces. DeepSeek has no confidence-matched set. Olmo's `f1k` is shown with the novel branch's sign; the familiar branch does not use it.

Model	Feature	Used as	Sign	Ours, first	Ours, matched	MAIN
Qwen	H_{4096}	S1/S2 readout	−	0.77	0.50	0.71
	`lble`	fallback ensemble	+	0.82	0.58	0.61
	`pmle`	fallback ensemble	+	0.80	0.65	0.50
	`jmp`	fallback ensemble	+	0.80	0.46	0.89
	`cfc`	fallback ensemble	−	0.69	0.48	0.43
	n			43 / 23	25 / 10	7 / 4
Skywork	`fa`	ensemble	+	0.68	0.29	0.75
	`chk`	ensemble	+	0.76	0.40	0.50
	`f1k`	ensemble	−	0.43	0.09	0.75
	n			19 / 12	16 / 8	4 / 3
Olmo	`f1k`	novel branch	+	0.39	0.64	1.00
	`chk`	familiar branch	+	0.72	0.45	0.50
	`lble`	familiar branch	+	0.56	0.58	0.50
	`pmle`	familiar branch	+	0.68	0.71	0.50
	seg-25% direction	S2 readout	+	0.99	0.39	1.00
	n			12 / 9	17 / 10	2 / 2
DeepSeek	`chk`	ensemble	+	0.71	—	0.67
	`nrk`	ensemble	−	0.63	—	1.00
	`cfl`	ensemble	−	0.65	—	0.62
	`pmle`	ensemble	+	0.68	—	0.05
	n			27 / 10	—	7 / 3

`f1k` and Olmo's novel-branch `f1k` points the shipped way. Both of those signs follow MAIN rather than our labels. On the held-out confidence-matched set, 8 of the 13 features are at or below 0.5, and Skywork's three features reverse (0.09–0.40), the pattern Section 6 describes. On MAIN, 10 of 17 are above 0.5, 5 are exactly 0.5 and 2 are below. DeepSeek's `pmle` reverses there (0.05), yet removing it costs 2 MAIN rows (Table 5): its value in the ensemble comes from how it combines with `nrk` and `cfl`, not from its own ranking. The Olmo direction's 0.99 on our first set is in-sample, because the direction was fitted there.

D Exploration: what else we tried

The submissions are the end of about 120 numbered experiments and 48 leaderboard builds (Appendix E). This appendix summarises the feature families we explored and why most were dropped, with emphasis on the white-box ones. Unless stated otherwise, "AUC" is the robust-vs-non-robust AUC on MAIN after controlling for trace length (robust = positive, so values below 0.5 mean the feature points the other way), and "our labels" means nested cross-validation (feature or layer selected inside each training fold) on the labelled sets of Section 2, reported as accuracy on forced / natural labels. Section 6 explains why results on our labels are weak evidence.

How the exploration went. We first screened features on a large pool of organiser-provided permutations, then found its non-robust labels were dominated by permutations that remove information from the problem, and moved to MAIN with length controls. Main's 43 rows could rank feature families but not validate ensembles: with the feature choice nested inside leave-one-problem-out, fixed mixes that looked near-perfect in-sample fell to 0.56–0.77. We then labelled our own problems to get more

rows, ran three GPU feature batteries over them (262–518 scalar features per trace, plus per-layer and per-head arrays), and from the per-model builds onward chose recipes with MAIN and leaderboard scores in view. The equal-weight percentile mean of Section 3 was never beaten held out by cascades, stacking or random forests; the last two fell below chance on MAIN.

Hidden states.

- *Prompt-token probe* (linear probe on the last prompt token, a few layers): below chance per model on MAIN (0.33). A ridge probe on the same state predicts thinking length (cross-validated R^2 0.57 on the organiser pool) but carries no robustness signal of its own.
- *Difference-of-means directions* per model and layer, fitted leaving one problem out: the right sign at every layer (0.57–0.67) but never beyond a permutation null. Mean states of fixed trace segments did better, 0.82 on MAIN for the segment at 25% of thinking; a difference-of-means direction on that segment, fitted on our labels, became S2's Olmo direction (Section 3.5).
- *Trajectory geometry* of the residual stream over time (tortuosity, speed, late-vs-early cosine, on 16-token waypoints and 16/64/256-token moving averages): mostly explained by length; late-vs-early cosine kept 0.78 after controls.
- *Anchoring*: cosine between the current reasoning state and the problem's state. Inverted on MAIN (0.30).
- *Step similarity*, prompted by work on self-reinforcing reasoning loops [Duan et al., 2026]: cosine between consecutive 64-token states. More similar steps go with non-robust (0.21–0.32 after averaging over chunk phases).
- *Reasoning as a state graph*, following the cluster-graph construction of Duan et al. [2026]: K-means states (K = 16, 32, 64) turn each trace into a walk; cosine kNN graphs on averaged trajectories. Robust traces form globally integrated graphs (kNN global efficiency 0.78; K-means cycle rank and spectral gap up to 0.86). The K-means features are at chance by the number of hits and vanish once step similarity is removed; the kNN signal survives that but holds in only three of the four models, and it is not stable across traces of the same problem: on MAIN, with one greedy and two temperature-1 traces per problem, its AUC is 0.82 / 0.50 / 0.45 and its rank correlation across the three traces is 0.06. Every leaderboard build that included it scored 0.28–0.50, except one in which it only broke ties.
- *Topological data analysis* [Tan et al., 2025] (Vietoris–Rips persistence on trajectory nodes; per layer and in sliding windows): at chance overall (7 of 214 tests at $p < 0.05$).
- *δ-hyperbolicity, compression and Poincaré embeddings* [Raj, 2026]: the best DeepSeek feature on our labels (δ at 50% depth, 0.81 / 0.79), but the hyperbolic map never beat plain geometry.
- *CKA* [Kornblith et al., 2019] across depth, and between clean and perturbed teacher-forced passes: weak; best Olmo cosine drift under code typos 0.79 / 0.76 on our labels.
- *Recall-vs-computation directions*, trained on separate recall and computation prompts: significant at the null rate on our labels.
- *Late-layer MLP write size* at the forced-answer position: larger writes go with non-robust on both MAIN (0.25) and our olympiad labels (0.31), consistent with a recalled answer injected late; it was not shipped.
- *Logit lens* [nostalgebraist, 2020] *on the answer digits*, following MemLens [He et al., 2025b] (where the answer digit emerges across layers, a memorisation signal): fully explained by whether the trace finished.

Attention.

- *Laplacian spectra of attention maps* [Bińkowski et al., 2025]: at or below chance as summary features on MAIN; as a trained probe, 0.73 on our olympiad labels and 0.83 when transferred to MAIN (22 rows). Selecting a single head inside each fold gave Olmo 0.92 / 0.85 on our labels; none of these was shipped.
- *Lookback Lens* [Chuang et al., 2024]: the paper's per-head logistic regression fails at our sample size (0.40–0.44 held out). Its late-minus-early summary became `lble`, and the same summary of prompt attention became `pmle`.

- *TOHA* [Bazarova et al., 2025] (minimum spanning forest from response tokens to the prompt, a faithful port with head selection nested): inverted on MAIN for the answer window (0.20); held out per model at most 0.66 (DeepSeek).
- *Attention periodicity and layer-to-layer change* (our own features: FFT over positions; KL between adjacent layers): nothing on MAIN.
- *Block jump*: attention-weighted backward distance between 64-token blocks, 0.70 on unfinished rows and not explained by looping or repetition (each $|\rho| \leq 0.14$). Shipped as `jmp`.

Interventions and perturbations.

- *Activation patching* [Meng et al., 2022] *and knockouts* (patching the no-reasoning run into the full trace; knocking out the problem tokens): the best features (0.67) duplicate forced-answer confidence and add nothing to it.
- *Rewording sensitivity* under three automatic rewordings, teacher-forcing the original trace: hidden-state cosine and log-likelihood change were shipped as `cfc` and `cfl`.
- *PEARL-style prose noise* [Djiré et al., 2025] (memorised inputs should break abruptly under rising input noise; we used letter typos at 2–30% in the prose only, never in numbers or math, and curves over the noise rate): mean-state drift area 0.77 on MAIN, but nothing in nested combinations.
- *A direct test*: regenerate the answer for five perturbations and call robust if all agree. Agreement AUC 0.81 for Skywork on our olympiad labels, but five extra long generations do not fit the per-call time limit.

Text and output distribution. Reasoning markers (hesitations, restarts) and the loop statistics of Duan et al. [2026] were null after length controls, except check share, which was shipped as `chk`. Sentence-embedding [Reimers and Gurevych, 2019] topic graphs reached 0.78 on MAIN (spectral gap; robust traces stay on fewer topics) but did not replicate on our 32k relabel. Forced-answer confidence and its slope over budgets were the strongest single features on MAIN (0.90 at 6144 tokens). Min-K% and Min-K%++ [Shi et al., 2024, Zhang et al., 2025] familiarity scores became the Olmo router.

Ports of recent papers. We also implemented seven recent methods as faithfully as our setting allows, checking each against its paper, and scored them on our earlier labelled unfinished rows (88 rows; 7% of tests at $p < 0.05$ against 5% by chance):

- *Layer-wise usable information* [Kim et al., 2024]: the strongest, 0.67 with the same direction in all four models, but correlated with forced-answer confidence and check share (ρ 0.41–0.57; 0.56 after removing them).
- *Local intrinsic dimension* [Yin et al., 2024]: 0.54 pooled, 0.77–0.80 for Skywork only.
- *LaRA* representation collapse [Gwak et al., 2026]: model-specific, from 0.08 to 0.65.
- *Attention rhythm* [Li et al., 2025], *receiver heads* from Thought Anchors [Bogdan et al., 2025], Min-K%++ [Zhang et al., 2025] and a circuit-guided internal–external discrepancy [Shen et al., 2026]: at chance. Thought Anchors' attention suppression pointed the wrong way (0.30) and not consistently across models.

E Builds scored on the leaderboard

Table 7: Leaderboard accuracy of our builds with their MAIN correct calls (each build's own recipe replayed on our MAIN traces). Per-model builds, the family S1 and S2 belong to, score highest on both.

Build	Leaderboard	MAIN (43)
v8-ent4096 (S1)	1.00	33
v8-ent4096-olmoseg25 (S2)	0.93	36
v8 (base recipe, no single-feature readouts)	0.93	32
v7, v5 (per-model, router added in v7)	0.93, 0.86	32, 34
v6 (per-model, before the Olmo router)	0.93	34
max (per-model)	0.86	33
lookback (per-model)	0.71	34
cascade, v4, ensemble (per-model)	0.64	30–34
single-family builds (24)	0.28–0.71	19–34

F Reproducibility

Submissions. S1 is our Submission 1 (`permodel-v8-ent4096-small.zip`, build `c03b14ac1abc`; 1.00 on the public leaderboard) and S2 is our Submission 2 (`permodel-v8-ent4096-olmoseg25-small.zip`, build `87f69d278fb3`). Both zips are in the repository with their SHA-256 hashes (README).

Code, recipes (`recipe.json`), shipped artifacts and the analysis scripts are at `https://anonymous.4open.science/r/AIMO-Interp-Challenge-Submission-Anonymised/`: `submission_core/` (inference), `analysis/build_submission_artifacts.py` (calibration: references, thresholds, per-model recipes), `analysis/build_datasets.py` (our labelled datasets, `datasets/`) and `reproduce/`. Running `python reproduce/replay.py --main <train-main-v2 parquet>` regenerates every MAIN number in Tables 1–4 and Appendices B–C on CPU in seconds. It uses the recipes inside the submitted zips, the shipped scoring code and our per-trace feature values (`reproduce/features.csv`); the README gives the details. The remaining analysis scripts are published for transparency and depend on intermediate data that is not released. Feature extraction and labelling ran on one NVIDIA RTX PRO 6000 and one A10G, using vLLM [Kwon et al., 2023] for labelling samples and HF Transformers for features. Model licences: Qwen3.5-4B, Skywork-OR1-Math-7B and Olmo-3-7B-Think are released under Apache-2.0, DeepSeek-R1-0528-Qwen3-8B under MIT. The challenge data are used under the competition's terms and are not redistributed.

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16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: An LLM (Claude Sonnet) wrote the perturbation texts for our own calibration labels, and a second open model (Qwen) acts as a judge for Olmo (Section 3); all LLM use is described in the AI usage statement.

Guidelines:

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